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PROJECT REPORT

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CUSINE PREDICTION AND NUTRITIONAL ANALYSIS OF GLOBAL RECIPES USING MACHINE LEARNING

Code link: Cuisine_Prediction_and_Nutritional_Analysis

ABSTRACT

Food recipes differ across regions in terms of their ingredients, preparation methods, and nutritional composition. Machine learning can be used to identify patterns in recipe ingredients and predict the cuisine to which a recipe may belong. Nutritional analysis can further help in understanding differences in the nutritional composition of recipes associated with different cuisines.

This project, titled “Cuisine Prediction and Nutritional Analysis of Global Recipes Using Machine Learning,” develops a cuisine classification system using ingredient information. The What's Cooking dataset, containing 39,774 recipes belonging to 20 cuisine categories, was used to train and evaluate two classification algorithms: Logistic Regression and Linear Support Vector Machine (Linear SVM). The ingredients were converted into numerical features using Term Frequency–Inverse Document Frequency (TF-IDF). The dataset was divided into training and testing sets in an 80:20 ratio. GridSearchCV with three-fold cross-validation was used for hyperparameter tuning.

The initial Logistic Regression model achieved 77.00% test accuracy, while the initial Linear SVM model achieved 77.16%. After hyperparameter tuning, Logistic Regression achieved 76.40% test accuracy, and Linear SVM retained 77.16% test accuracy. Therefore, Linear SVM was selected as the better-performing model based on the evaluated test accuracy.

For nutritional analysis, 50,000 recipes were sampled from the Food.com dataset, which originally contained 231,637 recipes. Their cuisines were predicted using the tuned Linear SVM model. Average calorie content and several nutrient-related Daily Value percentages were compared across the predicted cuisine groups using tables and visualizations.

The results demonstrate the usefulness of machine learning for recipe classification and exploratory nutritional analysis. However, the nutritional findings must be interpreted cautiously because the cuisine labels assigned to Food.com recipes are model predictions rather than verified ground-truth labels.

Keywords: Machine Learning, Cuisine Classification, TF-IDF, Logistic Regression, Linear SVM, GridSearchCV, Nutritional Analysis, Food Data.

1. INTRODUCTION

Food is an important part of cultural identity, and cuisines across the world differ in their ingredients and culinary traditions. For example, certain ingredient combinations are commonly associated with Indian, Italian, Mexican, Chinese, and other cuisines. Identifying such patterns manually can be difficult when working with thousands of recipes.

Machine learning provides techniques for discovering patterns in large datasets and using those patterns to make predictions. In recipe classification, a model can learn relationships between ingredient combinations and cuisine labels from a collection of labelled recipes. Once trained, the model can predict the likely cuisine of a new recipe based on its ingredients.

Apart from cuisine classification, nutritional information is useful for understanding the composition of food. Calories, fat, sugar, sodium, protein, saturated fat, and carbohydrates are important nutritional measures. Comparing these values across recipe groups can provide insights into the nutritional characteristics of a collection of recipes.

This project combines text-based machine learning with exploratory nutritional analysis. Two classification algorithms, Logistic Regression and Linear SVM, are trained and evaluated. The better-performing model is then applied to recipes from a second dataset, allowing nutritional values to be grouped and compared by predicted cuisine.

The project uses Python, Google Colab, Pandas, Scikit-learn, and Matplotlib to implement the complete workflow.

2. PROBLEM STATEMENT

Recipe datasets may contain thousands of recipes with different ingredients, names, and nutritional values. Manually identifying the cuisine of each recipe and comparing the nutritional composition of recipes across cuisine groups is time-consuming.

The problem addressed in this project is to develop a machine-learning system that predicts cuisine categories from recipe ingredients and uses those predicted categories to explore nutritional differences among recipes.

The project also compares two classification algorithms to determine which provides better predictive performance on an unseen test dataset.

3. OBJECTIVES

The main objectives of the project are:

1. To study and preprocess recipe datasets containing ingredients, cuisine labels, and nutritional information.
2. To convert ingredient text into numerical features using TF-IDF.
3. To develop a cuisine classification model using Logistic Regression.
4. To develop a second cuisine classification model using Linear SVM.
5. To evaluate both models using accuracy, weighted precision, weighted recall, and a confusion matrix.
6. To tune model hyperparameters using GridSearchCV and three-fold cross-validation.
7. To select the better-performing model based on the test results.
8. To predict cuisine categories for recipes in a separate dataset.
9. To compare average calories and nutrient-related Daily Value percentages across predicted cuisine groups.
10. To visualize the model performance and nutritional comparisons using charts.

4. DATASET DESCRIPTION

Two publicly available recipe datasets were used in this project.

4.1 What's Cooking Dataset

The What's Cooking dataset was obtained from the Kaggle What's Cooking competition. It contains recipes labelled with cuisine categories.

The important columns are:

- **id:** Unique identifier for a recipe.
- **cuisine:** Cuisine category associated with the recipe.
- **ingredients:** List of ingredients used in the recipe.

After preprocessing, the dataset contained 39,774 recipes and 20 distinct cuisine categories.

This dataset was used to train and evaluate the cuisine classification models because it provides known cuisine labels.

4.2 Food.com Recipes Dataset

The second dataset was obtained from Kaggle's Food.com Recipes and User Interactions dataset. The RAW_recipes.csv file contains recipe information, including names, ingredients, preparation details, and nutritional values.

Important fields include:

- **name:** Name of the recipe.

- **ingredients:** Ingredients used in the recipe.
- **nutrition:** A list containing seven nutritional values.
- **id:** Recipe identifier.

After removing duplicate recipe IDs and records with missing ingredients or nutrition information, the dataset contained 231,637 recipes.

A reproducible random sample of 50,000 recipes was selected for nutritional analysis.

Unlike the What's Cooking dataset, the Food.com recipe file used in this project does not provide reliable cuisine labels for the required classification task. Therefore, cuisine categories were predicted using the trained Linear SVM model.

4.3 Dataset Summary

Dataset	Purpose	Number of recipes
What's Cooking	Training and testing cuisine classifiers	39,774
Food.com	Nutritional analysis	231,637
Food.com sample	Cuisine prediction and nutritional comparison	50,000

5. TOOLS AND TECHNOLOGIES USED

The following tools and libraries were used:

Python: Programming language used for preprocessing, modelling, evaluation, and analysis.

Google Colab: Cloud-based notebook environment used to execute Python code.

Pandas: Used to load, clean, manipulate, and summarize recipe data.

Scikit-learn: Used for data splitting, TF-IDF feature extraction, classification, hyperparameter tuning, and evaluation.

Matplotlib: Used to create model comparison and nutritional analysis charts.

JSON and ZIP utilities: Used to load and extract the downloaded datasets.

Ast: Used to convert text representations of ingredient and nutrition lists into Python data structures where necessary.

6. METHODOLOGY

The project followed a sequence of data preprocessing, feature extraction, model training, evaluation, hyperparameter tuning, cuisine prediction, and nutritional analysis.

6.1 Data Loading

The downloaded datasets were extracted and loaded into Pandas DataFrames. The What's Cooking dataset was read from its JSON file, while the Food.com recipe data was read from RAW_recipes.csv.

The columns, missing values, and duplicate recipe identifiers were inspected to understand the structure and quality of the data.

6.2 Data Preprocessing

Preprocessing was performed to prepare the data for machine learning.

The following operations were carried out:

1. Duplicate recipe IDs were removed.
2. Records with missing cuisine labels or ingredient lists were removed from the classification dataset.
3. Recipes with empty ingredient lists were excluded.
4. Ingredient lists were converted into text strings.
5. Food.com ingredient lists stored as text were converted into actual Python lists.
6. Nutritional lists were converted into separate columns for analysis.

Each recipe in the What's Cooking dataset was represented as a single text string containing its ingredients. This representation allowed the text vectorizer to process the ingredient information.

6.3 Training and Testing Split

The What's Cooking dataset was divided into training and testing sets using an 80:20 split.

- Training data: approximately 31,819 recipes.
- Testing data: approximately 7,955 recipes.

The split used a random state of 42 to support reproducibility. Stratified splitting was applied to preserve the approximate distribution of cuisine categories in the two sets.

The training data was used to learn the model parameters, while the test data was reserved for evaluating predictions on unseen recipes.

6.4 Feature Extraction Using TF-IDF

Machine-learning algorithms require numerical input, but recipe ingredients are textual data. TF-IDF was used to transform ingredient text into numerical feature vectors.

Term Frequency–Inverse Document Frequency assigns importance to terms based on their occurrence in a document and their distribution across the collection of documents.

The TF-IDF representation can be expressed conceptually as:

TF-IDF(term, recipe) = Term Frequency × Inverse Document Frequency

Term frequency measures the importance of a term within a recipe, while inverse document frequency reduces the influence of terms that occur in many recipes.

The vectorizer used the following settings:

- Maximum features: 5,000.
- N-gram range: 1 to 2.
- Sublinear term frequency: Enabled.

The n-gram range allowed the model to use individual ingredient terms and pairs of terms. For example, an ingredient phrase such as “olive oil” could be represented alongside its individual words.

The vectorizer was fitted only on the training data. The same fitted vectorizer was then applied to the test data and the Food.com sample. This prevented the test data from influencing vocabulary construction.

7. MACHINE LEARNING ALGORITHMS

Two supervised classification algorithms were implemented.

7.1 Logistic Regression

Logistic Regression is a supervised learning algorithm commonly used for classification. Although its name contains the word regression, it can classify observations into multiple categories.

In this project, the model learned relationships between TF-IDF ingredient features and cuisine labels.

The multiclass Logistic Regression model estimated the cuisine category associated with a recipe based on its ingredient features.

The initial model used a maximum of 1,000 iterations and a random state of 42. The model was trained on the TF-IDF representation of the training recipes.

The trained model was then used to predict the cuisine categories of the test recipes.

The initial Logistic Regression model achieved **77.00% test accuracy**.

7.2 Linear Support Vector Machine

A Support Vector Machine is a supervised learning algorithm that identifies decision boundaries separating classes in feature space.

Linear SVM is suitable for high-dimensional text classification because it can operate effectively with sparse numerical feature representations such as TF-IDF.

In this project, Linear SVM learned boundaries separating the 20 cuisine categories using the ingredient features.

The model was trained using Scikit-learn's LinearSVC implementation, with a maximum of 3,000 iterations and a random state of 42.

The initial Linear SVM model achieved **77.16% test accuracy**.

7.3 Comparison of the Initial Models

The initial test results were:

Model	**Test accuracy**
Logistic Regression	77.00%
Linear SVM	77.16%

Linear SVM achieved slightly higher accuracy than Logistic Regression. Therefore, it was selected for further evaluation and subsequent cuisine prediction, subject to the results of hyperparameter tuning.

8. HYPERPARAMETER TUNING USING GRIDSEARCHCV

Hyperparameters are settings specified before model training. Their values can influence the predictive performance of a model.

GridSearchCV was used to identify suitable values of the regularization parameter C for both classification algorithms.

The parameter C controls the trade-off between model complexity and training errors in these regularized linear classifiers. Smaller values apply stronger regularization, while larger values apply weaker regularization.

8.1 Logistic Regression Tuning

The following C values were tested:

C = 0.1, 1, 10

Three-fold cross-validation was used. In each fold, the model was trained on part of the training data and validated on the remaining part.

The best parameter selected by GridSearchCV was:

Best C = 10

The best cross-validation accuracy was **76.71%**.

8.2 Linear SVM Tuning

The same candidate values were evaluated for Linear SVM:

C = 0.1, 1, 10

The best parameter selected by GridSearchCV was:

Best C = 1

The best cross-validation accuracy was **77.00%**.

8.3 Hyperparameter Tuning Results

Model	**Best C**	**Best cross-validation accuracy**
Logistic Regression	10	76.71%
Linear SVM	1	77.00%

Linear SVM achieved slightly higher cross-validation accuracy. However, cross-validation accuracy and test accuracy measure performance on different data partitions and should not be treated as identical.

9. MODEL EVALUATION

The tuned models were evaluated on the same held-out test dataset to compare their performance fairly.

The evaluation used accuracy, weighted precision, weighted recall, and a confusion matrix.

9.1 Accuracy

Accuracy measures the proportion of test predictions that match the actual cuisine labels.

Accuracy = Number of Correct Predictions / Total Number of Predictions

A higher accuracy indicates that the model correctly classifies a larger proportion of the test recipes.

9.2 Precision

Precision measures the proportion of predictions for a particular class that are correct.

Weighted precision combines the precision values of individual classes while accounting for the number of actual test samples belonging to each class.

9.3 Recall

Recall measures the proportion of actual instances of a class that are correctly identified.

Weighted recall combines recall across the cuisine categories, weighted by their support in the test dataset.

9.4 Confusion Matrix

A confusion matrix presents actual cuisine categories against predicted cuisine categories.

The diagonal entries represent correctly classified recipes. Entries outside the diagonal represent incorrect predictions, showing which cuisine categories the model confuses with one another.

The confusion matrix generated for the tuned Linear SVM showed strong diagonal counts for several cuisine categories, including Italian, Mexican, and Indian. Some recipes from related categories were incorrectly assigned to other cuisines.

This visualization helped identify both the successful classifications and the remaining errors.

9.5 Final Test Results

The actual results obtained from the Colab implementation were:

Model	Accuracy	Weighted precision	Weighted recall
Tuned Logistic Regression	76.40%	76.13%	76.40%
Tuned Linear SVM	77.16%	76.77%	77.16%

Linear SVM achieved the highest test accuracy of **77.16%** and the highest weighted precision of **76.77%**.

The weighted recall of Linear SVM was also 77.16%.

Although the improvement over Logistic Regression was small, Linear SVM performed better across the reported test metrics. It was therefore selected as the final classification model.

The results also show that hyperparameter tuning did not necessarily improve the test accuracy of every model. The tuned Logistic Regression model achieved 76.40%, compared with 77.00% for the initial model. This illustrates why model selection should consider validation results and independent test performance rather than assuming tuning always improves accuracy.

10. CUISINE PREDICTION ON THE FOOD.COM DATASET

After evaluating the classifiers, the tuned Linear SVM model was applied to a sample of 50,000 Food.com recipes.

The ingredient lists were converted into text strings and transformed using the same TF-IDF vectorizer fitted on the What's Cooking training data. The tuned Linear SVM then predicted a cuisine category for each recipe.

The predictions were stored in a new column called predicted_cuisine.

All 50,000 sampled recipes received a predicted cuisine category.

10.1 Distribution of Predicted Cuisines

The five most frequent predicted categories were:

Predicted cuisine	Number of recipes
Southern US	12,980
Italian	11,111
Mexican	6,207
French	4,031
Chinese	1,839

The model assigned recipes to all 20 cuisine categories. Southern US, Italian, and Mexican were the three largest groups in the selected sample.

These counts describe the model's predictions, not the verified cuisine distribution of the Food.com dataset. The model may assign a recipe to a cuisine even when the recipe is not traditionally associated with that cuisine.

The predicted distribution may also reflect differences between the two datasets, including differences in ingredient descriptions and recipe collection methods.

11. NUTRITIONAL ANALYSIS

The second major component of the project was to compare nutritional characteristics across the predicted cuisine categories.

The Food.com dataset contained a nutrition field represented as a list of seven values. These values were separated into individual columns to support analysis.

The nutritional variables used were:

1. Calories
2. Total fat Daily Value percentage
3. Sugar Daily Value percentage

4. Sodium Daily Value percentage
5. Protein Daily Value percentage
6. Saturated fat Daily Value percentage
7. Carbohydrates Daily Value percentage

The average value of each nutritional variable was calculated for the 50,000 sampled recipes, grouped by predicted cuisine.

11.1 Average Calories by Predicted Cuisine

A bar chart was created to compare the average calorie value across the predicted cuisine categories.

The chart provided a visual comparison of the average calorie content of recipes in each group and made it easier to identify groups with relatively higher or lower mean calorie values.

The results represent average calories per recipe, not calories per serving. Recipe serving sizes can differ considerably, so these values should not be interpreted as direct measures of the energy content of a typical serving.

11.2 Protein and Sugar Comparison

A grouped bar chart was generated to compare the average protein and sugar Daily Value percentages across predicted cuisines.

This comparison illustrated differences in the recorded protein and sugar values among the cuisine groups. A higher protein Daily Value percentage indicates a greater recorded contribution to the reference Daily Value, while a higher sugar percentage indicates a greater recorded sugar Daily Value.

The values are percentages of Daily Value, not grams of protein or sugar.

11.3 Sodium and Total Fat Comparison

A further grouped bar chart compared the average sodium and total fat Daily Value percentages across predicted cuisine categories.

This visualization supported an exploratory comparison of the recorded sodium and fat values among the groups.

Higher values indicate higher average recorded Daily Value percentages. However, these values should not be used to conclude that an entire cuisine is inherently healthier or less healthy than another cuisine.

11.4 Interpretation of Nutritional Results

The nutritional analysis demonstrates how machine learning predictions can be combined with nutritional information to group and summarize a large recipe collection.

The approach makes it possible to compare average nutritional values across automatically generated cuisine categories without manually labelling every recipe.

However, the interpretation is subject to several limitations. First, cuisine labels for the Food.com recipes were predicted rather than verified. Second, the two datasets may describe ingredients differently, which can affect the classifier. Third, average values per recipe do not account for differences in serving size. Finally, nutritional values stored as percentages of Daily Value should not be confused with measurements in grams.

Therefore, the nutritional charts should be understood as exploratory summaries of the sampled recipes grouped by predicted cuisine.

12. RESULTS AND DISCUSSION

The project successfully implemented a machine-learning pipeline for cuisine classification and exploratory nutritional analysis.

The What's Cooking dataset provided 39,774 labelled recipes across 20 cuisine categories. TF-IDF transformed the ingredient text into numerical features suitable for the classification algorithms.

Both Logistic Regression and Linear SVM achieved test accuracies of approximately 77%. The initial Logistic Regression model achieved 77.00% accuracy, while the initial Linear SVM achieved 77.16%.

After hyperparameter tuning using three-fold cross-validation, Logistic Regression selected C = 10 and Linear SVM selected C = 1. The tuned Linear SVM achieved the highest test accuracy of 77.16%, weighted precision of 76.77%, and weighted recall of 77.16%.

The tuned Linear SVM was subsequently applied to a reproducible sample of 50,000 Food.com recipes. All 20 cuisine categories were represented among the predictions, although some categories were much more frequent than others.

The nutritional analysis produced grouped averages for calories, fat, sugar, sodium, protein, saturated fat, and carbohydrates. Visualizations were created to compare calories, protein and sugar, and sodium and total fat.

Overall, the project demonstrates that ingredient-based text classification can be combined with nutritional data to conduct exploratory analysis on a large collection of recipes. The performance results also show that the choice of algorithm and hyperparameters affects classification outcomes.

13. LIMITATIONS

Although the project achieved its main objectives, several limitations should be acknowledged.

13.1 Predicted Cuisine Labels

The Food.com dataset used for nutritional analysis did not provide reliable ground-truth cuisine labels for this task. Therefore, the trained model assigned the cuisine categories.

The nutritional comparisons are consequently based on predicted labels and may contain classification errors.

13.2 Differences Between Datasets

The What's Cooking and Food.com datasets were collected independently. They may differ in recipe styles, ingredient naming conventions, and data distributions.

This domain difference can reduce the reliability of predictions on Food.com recipes compared with predictions on the original test dataset.

13.3 Moderate Classification Accuracy

The best test accuracy was 77.16%. This indicates useful predictive performance but also means that some recipes were incorrectly classified.

Recipes containing ingredients common to multiple cuisines may be especially difficult to classify.

13.4 Nutritional Measurements

The non-calorie nutritional values used in the analysis are percentages of Daily Value, not absolute quantities in grams or milligrams.

The averages were calculated per recipe rather than per standardized serving. Therefore, comparisons may be influenced by differences in recipe yield and serving size.

13.5 Limited Hyperparameter Search

Only three values of C were tested for each classifier. Additional parameter values, alternative vectorization settings, or other algorithms might produce different results.

13.6 No Ground-Truth Nutritional Group Validation

Because the Food.com cuisine labels were not independently verified, the project could not directly evaluate the accuracy of the nutritional grouping against confirmed cuisine categories.

Future work could address this limitation by obtaining a dataset with verified cuisine labels and standardized nutritional values.

14. FUTURE SCOPE

The project can be extended in several ways.

1. **Larger hyperparameter search:** More values of C and other model settings could be evaluated to improve classification performance.
2. **Additional algorithms:** Models such as Multinomial Naive Bayes, Random Forest, and gradient-boosting methods could be compared with the current classifiers where appropriate.
3. **Verified cuisine labels:** A labelled dataset containing both ingredients and nutritional information would support more reliable nutritional comparisons.
4. **Improved ingredient processing:** Ingredient normalization, handling of synonyms, and more advanced text representations could improve predictions.
5. **Standardized serving sizes:** Nutritional values could be compared per serving or per 100 grams to improve the fairness of nutritional comparisons.
6. **Interactive application:** The trained model could be integrated into a simple web application where users enter a list of ingredients and receive a predicted cuisine category.
7. **Recipe recommendation:** The classification system could be extended to suggest recipes based on ingredients available to a user.
8. **Nutritional filtering:** A future application could allow users to filter recipes based on nutritional attributes, provided the underlying nutrition data is sufficiently reliable.

15. CONCLUSION

The project “Cuisine Prediction and Nutritional Analysis of Global Recipes Using Machine Learning” successfully demonstrated the use of text-based machine learning for recipe classification and exploratory nutritional analysis.

The What's Cooking dataset was preprocessed, and recipe ingredients were transformed into numerical features using TF-IDF. Logistic Regression and Linear SVM were trained and evaluated on a held-out test set. GridSearchCV with three-fold cross-validation was used to tune the regularization parameter C for both models.

The tuned Linear SVM performed best among the evaluated models, achieving 77.16% test accuracy, 76.77% weighted precision, and 77.16% weighted recall. It was then used to predict cuisine categories for 50,000 Food.com recipes.

The nutritional analysis grouped these recipes by predicted cuisine and compared average calories, fat, sugar, sodium, protein, saturated fat, and carbohydrates. The results were presented using tables and visualizations to make the nutritional patterns easier to examine.

The project illustrates how machine learning can extract patterns from ingredient text and how predictions can be combined with structured nutritional information for further analysis. At the same time, the findings emphasize the importance of verified labels, consistent nutritional measurements, and careful interpretation when working with multiple datasets.

In conclusion, the project achieved its main objectives and provides a foundation for further work in food classification, recipe analysis, and nutrition-oriented applications.

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